Deep Time-Series Models under Observation Shift

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Deep learning offers a wide range of architectures for time-series modelling, from recurrent neural networks (RNNs) and Transformers to models specifically designed for irregularly sampled data. Despite increasingly large benchmarking efforts, there is no universally best architecture: relative performance can vary substantially across datasets and temporal patterns. Recent benchmarks have begun to move beyond aggregate rankings towards understanding how model performance relates to structural properties of the underlying time series [1, 2].

One potentially important but less well-understood aspect is the observation process. In many real-world datasets, and particularly in healthcare, observations are neither regular nor randomly missing. Measurements may be collected because the underlying state has changed: for example, a doctor may order lab tests more often for a deteriorating patient. Missingness and sampling patterns can themselves contain predictive information. Some models, e.g. GRU-D, explicitly exploit this phenomenon by incorporating missingness masks and elapsed time into their recurrent dynamics [3].

This creates an interesting tension. A model that exploits the observation process may achieve better prediction when training and test data are collected in the same way, while becoming brittle if the observation policy changes. Conversely, architectures designed to model continuous underlying dynamics may respond differently to such changes. Neural Controlled Differential Equations (Neural CDEs), for example, were developed specifically for partially observed, irregularly sampled multivariate time series and model a continuous hidden state driven by the observed path [4].

Question: When do inductive biases designed for irregular time series improve prediction, and are these advantages robust when the observation process changes?

Rather than searching for a universally best model, the goal is to identify which aspects of temporal data different architectures exploit, and under which conditions these advantages remain reliable. The longer-term goal is to move from understanding which time-series models 'win' in traditional benchmarks or leaderboards towards understanding what the model has actually learned from the temporal data, and whether that information remains useful when the data are collected differently.

Models

A small set of architectures can be selected to represent meaningfully different approaches to temporal modelling:

- GRU or LSTM: a standard recurrent architecture, providing a baseline without specialised treatment of irregular observations;
- GRU-D: an extension of recurrent modelling that explicitly incorporates missingness masks, time gaps, and decay, and is therefore designed to exploit informative missingness [3];

- Neural CDE: a continuous-time model designed for partially observed and irregularly sampled sequences [4];
- Transformer: an attention-based alternative, augmented as appropriate with representations of observation time and missingness.

Modern state-space or continuous-time state-space models may provide an additional extension, but a careful comparison of three or four architectures is preferable to a large model leaderboard.

Experimental setup

A central methodological consideration is to distinguish architectural effects from differences in available information. If one model receives observation masks and time gaps while another receives only measured values, differences in performance cannot straightforwardly be attributed to architecture. Input ablations–e.g., values only vs values plus time gaps and missingness indicators–can therefore form part of this analysis.

Synthetic or semi-synthetic data are particularly useful because temporal dynamics and observation mechanisms can be manipulated independently. Experiments may subsequently be extended to realistic datasets such as MIMIC-IV [5], electricity-demand data, or other public longitudinal/time-series datasets that allow for testing different regimes.

Potential research directions

1. When does modelling irregularity help?
Starting from a known temporal process, the sampling density and irregularity can be systematically varied while keeping the underlying dynamics and prediction task fixed. This provides a controlled test of whether models designed for irregular data actually become more advantageous as observations become sparser or less regular. Missingness can initially be introduced independently of the underlying state, providing a baseline before considering informative observation processes.

The comparison can additionally separate the effect of architecture from the effect of explicitly providing timestamps, time gaps, and missingness indicators.

2. When is the observation process itself useful?
The observation mechanism can then be made informative. For example, the probability of observing a variable may increase when an underlying latent state crosses a threshold or moves towards an adverse outcome. This creates a setting in which observation patterns genuinely contain predictive information. Models such as GRU-D are explicitly designed to exploit such patterns [3]. Then, resulting experiments can examine:

- whether informative observation improves prediction;
- which architectures benefit most from incorporating this;
- whether models use measured values and observation patterns differently;
- how performance changes when masks or timing information are removed.

3. Are these advantages robust to observation-process shift?
Finally, the underlying temporal process can be held fixed while the observation mechanism changes between training and testing:

$$P_{\text{train}}(O_t \mid Z_t) \neq P_{\text{test}}(O_t \mid Z_t),$$

where Z_t denotes the underlying state and O_t whether it is observed.

For example, measurements could be strongly state-dependent during training but less state-dependent at test time, or two environments could follow different monitoring policies. This setting provides a controlled way to test whether strong in-distribution performance results from learning robust temporal relationships or from exploiting an observation-specific shortcut. A particularly interesting possibility is that explicitly modelling informative missingness improves in-distribution prediction while increasing sensitivity to changes in the observation mechanism. The aim is not to assume that such a trade-off exists, but to establish experimentally when it does or does not arise.

Possible project directions

A feasible project could focus on three architectures and one carefully controlled family of observation mechanisms. The emphasis should be on experimental design and analysis rather than exhaustive benchmarking. Multiple random seeds, consistent data splits and preprocessing, and controlled and correct model setup and hyperparameter tuning should be used.

You aim to support a statement about a regime of model behaviour, rather than just report that one architecture performs best on one dataset. For example, a possible finding might be that explicitly modelling missingness improves prediction when observation patterns are stable and informative, but this advantage decreases when the observation policy changes. This is an illustrative hypothesis rather than an expected result. A demonstration that the anticipated effect does not occur, or that different architectures respond similarly, would also be a meaningful outcome.

Extensions could include one or more real datasets from an application field of your choice, additional types of temporal structure, representation-level analysis, robustness methods that discourage reliance on unstable observation patterns, or modern continuous-time/state-space architectures. A further direction would be to ask whether measurable characteristics of a dataset can predict which modelling assumptions are likely to be beneficial. Recent work has begun to investigate this broader model-selection question at scale [1, 2], making the controlled study of why particular inductive biases succeed a complementary direction.

References

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